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Inventor(s)	Ribeiro; Claudio Santiago et al.

Transformer monitoring apparatus

Abstract

A transformer monitor includes a housing and at least two connectors. The housing has a first end and a second end opposite the first end. A first connector is positioned proximate the housing's first end and configured for connection to a first cable assembly that outputs signals representative of a primary input current to the transformer. A second connector is positioned proximate the housing's second end and configured for connection to a second cable assembly that outputs signals representative of a secondary output current of the transformer. The connectors are oriented such that the first cable assembly extends away from the housing's first end in a first direction when the cable assembly is connected to the first connector and the second cable assembly extends away from the housing's second end in a second direction generally opposite to the first direction when the cable assembly is connected to the second connector.

Inventors: Ribeiro; Claudio Santiago (Evanston, IL), Murray; Elliott Earl (Plantation, FL), Claeys; Patrick Stuart (Weston, FL)

Applicant: UBICQUIA, INC. (Fort Lauderdale, FL)

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Primary Examiner: Nguyen; Hoai-An D.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation-in-part of U.S. Design patent application Ser. No. 29/830,655, which was filed on Mar. 14, 2022, and is incorporated herein by this reference as if fully set forth herein. Additionally, the present application claims the benefit of and priority upon U.S. Provisional Patent Application

TECHNICAL FIELD

(1) The present disclosure relates generally to electrical utility monitoring equipment, and, more particularly, to a transformer monitoring apparatus that is configured to sense current on both the primary and secondary sides of a transformer while mitigating interference or crosstalk between current sensing cable assemblies.

BACKGROUND

(2) Electrical utility transformers are used extensively for converting distribution voltage levels down to service levels that are provided to end customers. Transformers are commonly provided as pole-mounted units on utility poles and as pad mounted units where electrical utility lines are underground. Transformers are subject to failures, which then cause service outage for customers being serviced by the transformer.

(3) Oftentimes, a utility may not know when a transformer has failed until an outage is reported. Accordingly, there is a need for a utility to be able to monitor its distribution transformers to identify problems before they cause an outage, as well as to identify when an outage has occurred. To accurately assess the condition or health of a transformer, parameter data associated with the primary and secondary sides of the transformer are needed. However, the accuracy of such data may be negatively impacted due to interference or crosstalk between cable assemblies used to provide output signals associated with the primary and secondary sides of the transformer.

SUMMARY

(4) According to one exemplary embodiment of the present disclosure, a monitoring apparatus for a transformer includes a housing and at least two connectors. The housing has a first end and a second end, where the second end is generally opposite the first end. The two ends may be lengthwise ends, widthwise ends, or angular ends. The housing may also have multiple faces or surfaces on a front side thereof, which may be on different planes. A first one of the connectors is positioned proximate the first end of the housing and configured for connection to a first cable assembly that outputs signals representative of at least one primary input current to the transformer. A second one of the connectors is positioned proximate the second end of the housing and configured for connection to a second cable assembly that outputs signals representative of at least one secondary output current of the transformer. The first connector and the second connector are oriented such that the first cable assembly extends away from the first end of housing in a first direction when the first cable assembly is connected to the first connector and the second cable assembly extends away from the second end of the housing in a second direction when the second cable assembly is connected to the second connector, and wherein the first direction and the second direction are generally opposite directions.

(5) According to another exemplary embodiment of the present disclosure, the first connector is oriented at an angle of not more than forty-five degrees relative to a reference plane running a length of the housing, and the second connector is oriented at an angle of not less than one hundred thirty-five degrees relative to the reference plane. Alternatively, the second connector is oriented at an angle of not more than forty-five degrees relative to a reference plane running a length of the housing, and the first connector is oriented at an angle of not less than one hundred thirty-five degrees relative to the reference plane. Further alternatively, the first connector and the second connector may be oriented so as to be at an angle in a range of about ninety degrees to about one hundred eighty degrees relative to each other.

(6) According to another exemplary embodiment of the present disclosure, each of the first connector and the second connector may include a respective connector component that extends outward from the housing. Alternatively or additionally, the first cable assembly may output signals representative of at least two primary input currents to the transformer and/or the second cable

assembly may output signals representative of at least two secondary output currents of the transformer. Further alternatively or additionally, the first connector and the second connector may be coaxial connectors. Still further, where the housing has a front side and a back side, the first connector and the second connector may be positioned on the front side of the housing.

(7) According to another exemplary embodiment of the present disclosure, the housing may include a base on a back side thereof, wherein the base defines a cavity. In this embodiment, the monitoring apparatus may further include a pressure sensor connector and an antenna connector positioned in the cavity. Additionally, a wall of the base may define at least two cutouts to allow for passage of an antenna cable to the antenna connector and a pressure sensor cable to the pressure sensor connector.

(8) According to a further exemplary embodiment of the present disclosure, the monitoring apparatus may additionally include a third connector positioned proximate the second end of the housing. In such a case, the third connector may be configured for connection to a third cable assembly that outputs alternating current (AC) power for use by the monitoring apparatus. Alternatively or additionally, the third connector may be oriented substantially parallel to the second connector.

(9) According to yet another exemplary embodiment of the present disclosure, a monitoring apparatus for a transformer includes a housing and at least three connectors. According to this embodiment, the housing has a front side, a back side, a first lengthwise end, and a second lengthwise end, where the second lengthwise end is substantially opposite the first lengthwise end. A first one of the connectors is positioned on the front side of the housing proximate the first lengthwise end and configured for connection to a first cable assembly that outputs signals representative of at least one primary input current to the transformer. A second one of the connectors is positioned on the front side of the housing proximate the second lengthwise end and configured for connection to a second cable assembly that outputs signals representative of at least one secondary output current of the transformer. A third one of the connectors is positioned on the front side of the housing proximate the second lengthwise end and configured for connection to a third cable assembly that outputs AC power for use by the monitoring apparatus. In this embodiment, the first connector and the second connector are oriented such that the first cable assembly extends away from the first end of housing in a first direction when the first cable assembly is connected to the first connector, the second cable assembly extends away from the second end of the housing in a second direction when the second cable assembly is connected to the second connector, and the first connector and the second connector are oriented so as to be at an angle in a range of about ninety degrees to about one hundred eighty degrees relative to each other.

(10) Positioning the connectors for the current sensing cable assemblies on different faces or at different ends of the monitoring apparatus mitigates the likelihood that the primary side and secondary side current sensing cable assemblies will cross each other, which could cause errors in the primary side and/or secondary side current data computed by a processor based on outputs of the cable assemblies.

(11) Although the present disclosure illustrates and describes a transformer monitoring apparatus, it is nevertheless not intended to be limited to the details shown because various modifications and structural changes may be made therein without departing from the spirit of the disclosure and while remaining within the scope and range of equivalents of the claims. Additionally, well-known elements of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

(12) Features that are considered characteristic of the invention are set forth in the appended claims. As required, detailed embodiments of the transformer monitoring apparatus are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary, and the transformer monitoring apparatus may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as

a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the claimed invention in appropriately detailed structures. Further, the terms and phrases used herein are not intended to be limiting; but rather provide an understandable description of the disclosure. While the specification concludes with claims defining the features of the invention, it is believed that the claimed invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. The figures of the drawings are not drawn to scale.

(13) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein mean “included, but not limited to.” The term “providing” is defined herein in its broadest sense (e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time).

(14) As used in this description, unless otherwise specified, azimuth or positional relationships indicated by terms such as “up”, “down”, “left”, “right”, “inside”, “outside”, “front”, “back”, “head”, “tail” and so on, are azimuth or positional relationships based on the drawings, which are only to facilitate description of the embodiments of the present disclosure and simplify the description, but not to indicate or imply that the devices or components must have a specific azimuth, or be constructed or operated in the specific azimuth, which thus cannot be understood as a limitation to the embodiments of the present disclosure or the appended claims. Furthermore, terms such as “first”, “second”, “third” and so on are only used for identification purposes and are not to be construed as indicating or implying relative importance.

(15) As used in this description, unless otherwise clearly defined and limited, terms such as “installed”, “coupled”, and “connected” should be broadly interpreted, for example, to mean fixedly connected, detachably connected, integrally connected; mechanically connected, electrically connected; directly connected or indirectly connected via an intermediate medium.

(16) In the absence of any specific clarification related to its express use in a particular context, where the terms “substantially,” “approximately,” “generally,” or “about” are used as modifiers in the present disclosure and any appended claims (e.g., to modify a structure, a dimension, a measurement, a direction, an orientation, or any other characteristic), it is understood that the characteristic may vary by up to thirty percent. In these cases, a device that is mounted exactly orthogonal is mounted along a “Y” axis and a “X” axis that is normal (i.e., 90 degrees or at right angle) to a plane or line formed by a “Z” axis. Different from the exact precision of the term “orthogonal,” the use of “substantially” or “about” to modify the characteristic permits a variance of the particular characteristic by up to thirty percent or such other amount as may be required by the context of the applicable description. As another example, an element or component that is oriented exactly opposite to another element or component is oriented one hundred eighty degrees from the other element or component. Different from the exact precision of the term “opposite,” the use of “substantially” or “generally” to modify the orientation or direction permits a variance thereof by up to thirty percent (e.g., up to fifty-four degrees) or such other amount as may be stated in the applicable description or required by the context of the applicable description. Alternatively, these terms may refer to a range of numbers or quantities that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances, these terms may include numbers that are rounded to the nearest significant figure.

(17) In the present disclosure, the term “longitudinal” or “lengthwise” means in a direction corresponding to an elongated direction of the device. Those skilled in the art will understand the specific meanings of the above-mentioned terms and others in the embodiments of the present disclosure according to the described context, examples, and circumstances.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and explain various principles and advantages all in accordance with the present disclosure.
- (2) FIG. 1 is a front side perspective view of a transformer monitoring unit in accordance with an exemplary embodiment of the present disclosure.
- (3) FIG. 2 is a side elevational view of the transformer monitoring unit of FIG. 1.
- (4) FIG. 3 is a back side perspective view of the transformer monitoring unit of FIG. 1.
- (5) FIG. 4 is a front side perspective view of the transformer monitoring unit of FIG. 1 showing connection details between cable assemblies and connectors of the transformer monitoring unit, in accordance with exemplary embodiments of the present disclosure.
- (6) FIG. 5 is a front perspective view of the transformer monitoring unit of FIG. 1 as connected to various cable assemblies for use in monitoring an aerial pole-mounted transformer, in accordance with additional exemplary embodiments of the present disclosure.
- (7) FIG. 6 illustrates a use case of the transformer monitoring unit of FIG. 1 as mounted to an aerial distribution transformer, in accordance with further exemplary embodiments of the present disclosure.
- (8) FIG. 7 is a front perspective view of the transformer monitoring unit of FIG. 1 as connected to various cable assemblies for use in monitoring a pad mounted transformer, in accordance with further exemplary embodiments of the present disclosure.
- (9) FIG. 8 illustrates a use case of the transformer monitoring unit of FIG. 1 as mounted to a pad mounted distribution transformer, in accordance with further exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

- (10) In order to address problems associated with using a field-retrofittable monitor to accurately detect primary current(s) into and secondary current(s) out of a utility transformer, a transformer monitoring unit or apparatus is disclosed in accordance with several exemplary embodiments. The disclosed transformer monitoring unit is configured to maintain physical separation between a primary current-sensing cable assembly and a secondary current-sensing cable assembly when the transformer monitoring unit and its attached cable assemblies are mounted in a retrofit manner to a transformer in the field.
- (11) FIG. 1 is a front side perspective view of a transformer monitoring apparatus or unit **100** in accordance with some exemplary embodiments of the present disclosure. The transformer monitoring unit **100** is usable to monitor a single-phase or a three-phase transformer and includes a housing **102** that is generally cuboid in shape, having a front side **104**, lateral sides **106**, and a back side **300** (shown in FIG. 3). In the exemplary orientation shown in FIG. 1, the housing **102** is elongated in the vertical direction, although need not be. The housing **102** has a first end **110** and a second end **108** at generally opposite lengthwise ends of the housing **102**. Alternatively, the generally opposing ends of the housing **102** may be along the lateral sides **106** of the housing **102** (e.g., widthwise) or at generally opposite faces of a faceted housing. In the embodiment depicted in FIG. 1, a distance between the first and second ends **108**, **110** is greater than the width of the housing **102**, which is in the horizontal direction as shown in FIG. 1. The front side **104** of the housing **102** is generally flat along a vertical plane and becomes rounded towards each of the first and second ends **108**, **110** towards the rear or back of the transformer monitoring unit **100** where a base **112** is mounted or included.

(12) A first connector **124** is positioned on the front side **104** of the housing **102** proximate the first end **110** of the housing **102**. The first connector **124** may have a metal connector component that includes a cylindrical ground portion surrounding several signal conductors, and which extends outwardly from the front side **104** of the housing **102**. The first connector **124** may be supported by a boss or lip **122** formed proximate the first end **110** of the housing **102**. The first connector **124** may be a multi-pin coaxial connector configured for connection to a cable assembly that outputs signals representative of at least one primary input current to a transformer being monitored. The cable assembly may include one or more Rogowski coils or current transformers for sensing current at the primary side of the transformer. The cable assembly may include two or more Rogowski coils or current transformers for sensing at least two input currents at the primary side of the transformer when the transformer is a three-phase transformer.

(13) Similarly, a second connector **118** is positioned on the front side **104** of the housing **102** proximate the second end **108** of the housing **102**. The second connector **118** may have a metal connector component that includes a cylindrical ground portion surrounding several signal conductors, and which extends outwardly from the front side **104** of the housing **102**. The second connector **118** may be supported by a boss or lip **114** formed proximate the second end **108** of the housing **102**. The second connector **118** may be a multi-pin coaxial connector configured for connection to a cable assembly that outputs signals representative of at least one secondary output current of a transformer being monitored. The cable assembly may include one or more Rogowski coils or current transformers for sensing current at the secondary side of the transformer. Where the transformer includes multiple secondary outputs, the cable assembly may include two or more Rogowski coils or current transformers for sensing the multiple output currents supplied by the secondary side of the transformer.

(14) The monitoring unit **100** may optionally include a third connector **120** positioned on the front side **104** or along a lateral wall **106** of the housing **102**. According to one embodiment, the third connector **120** may be positioned proximate the second end **108** of the housing **102**, as illustrated in FIG. 1. When positioned proximate the second end **108** of the housing **102**, the third connector **120** may be oriented to be substantially parallel to the second connector **118**. The third connector **120** may be supported by a boss or lip **114** formed at its position on the housing **102**. When included, the third connector may be configured for connection to a third cable assembly that outputs AC power for use by the monitoring unit **100**. In a preferred embodiment, the AC power supplied through the third connector **120** is supplied at the voltage level at the secondary side of the transformer (e.g., 120 VAC) and thereby permits sensing of such voltage by the monitoring unit **100** in accordance with known voltage sensing techniques. Each connector **118**, **120**, **124** may include mechanical elements or features to permit secure connection to a mating connector of a cable assembly, such as, for example, one or more lugs, nubs, or posts that extend perpendicularly outward from the connector component or external or internal threads.

(15) FIG. 2 is a side elevational view of the exemplary transformer monitoring unit **100** as viewed looking toward the right lateral side **106**. This view shows that the surface of the front side **104** rounds off or tapers as it approaches the first and second ends **108**, **110** of the housing **102** and includes rounded corner portions **212**, **214**, respectively. As shown in FIG. 2, the ends **108**, **110** of the housing **102** define a plane along line **202**. Each of the first and second connectors **118**, **124** extend from the front side **104** of the housing **102** proximate a respective end **108**, **110** and along a respective axis **204**, **208**. In this embodiment, the third connector **120** is parallel to the second connector **118** and, therefore, not shown in the side view of FIG. 2. Axis **204** and axis **206**, indicating the orientations of the first and second (and third) connectors **118**, **124**, illustrate that the connectors **118**, **124** may be oriented so as to be at a non-zero angle relative to reference plane **202**. In general, the angles of orientation of the first and second connectors **118**, **124** are such that the first and second connectors **124**, **118** are generally oriented in opposing directions, but less than one hundred eighty degrees with respect to each other. In some exemplary embodiments, the angle

between reference plane **202** and each axis **204**, **206** may be forty-five degrees or less. Thus, the terms “generally opposite,” “substantially opposite,” “generally opposite direction(s),” “generally opposite orientation(s),” “substantially opposite direction(s),” and “substantially opposite orientation(s),” mean an angle between axis **204** and axis **206** that is greater than or equal to ninety degrees. In the embodiment illustrated in FIG. 2, the angle between axis **204** and axis **206** is approximately one hundred thirty-five degrees.

(16) By orienting the first and second connectors **118**, **124** in such a manner, cables of the cable assemblies connected to these connectors **118**, **124** extend away from the connectors **124**, **118** and the front side **104** of the housing **102** in generally opposing directions, providing clearance for an operator to connect or remove the cables and mitigating the likelihood that such cables will be crossed during use. Forcing the cables to extend away from each other helps to ensure that the primary and secondary side sensing cables are separated, thereby mitigating the likelihood of an electric field generated around either cable negatively affecting the other. In some embodiments, the housing **102** can extend forward sufficiently such that the first and second connectors **118**, **124** can be oriented in directly opposite directions, at one hundred eighty degrees relative to each other.

(17) On the back side of the monitoring unit **100**, a base **112** may be attached or form part of the housing **102**. The base **112** may extend from the reference plane **202** and facilitate mounting of the monitoring unit **100** to a transformer surface. In some embodiments, magnets **208**, **210** may be included on the back side of the housing **102** and extend beyond the edge of the base **112** to magnetically engage the transformer surface, which is typically made of a ferrous metal. As a result, the magnets **208**, **210** may hold the transformer monitoring unit **100** in place during use.

(18) FIG. 3 is a back side perspective view of the exemplary transformer monitoring unit **100**. As shown in this view, the base **112** may form a wall around a perimeter of the back side **300** of the transformer monitoring unit **100**. Magnets **208A**, **208B**, **210A**, **210B** may be positioned generally along the lengthwise sides of the back side **300** of the housing **102**. A recess or cavity **302** may be formed into the back side **300** of the housing **102** to provide a space in which an antenna connector **304** and a pressure sensor connector **306** may be included. When included, the antenna connector **304** connects a radio module of the monitoring unit **100** to an external antenna via an antenna cable to allow the transformer monitoring unit **100** to communicate with a remote station. The radio module may be, for example, a cellular data unit that is capable of connecting to a cellular communications infrastructure. When included, the pressure sensor connector **306** connects to a pressure sensor via a pressure sensor cable **622**, and the pressure sensor provides an electrical signal that indicates the pressure inside the oil tank of the transformer. To allow the cables to exit the base **112**, the wall of the base **112** defines a pair of cutouts **308A**, **308B** or notches to allow the antenna cable (not shown) and the pressure sensor cable **622** to pass to the antenna connector **304** and the pressure sensor connector **306**, respectively, when the transformer monitoring unit **100** is mounted to a transformer.

(19) FIG. 4 is a front side perspective view of the exemplary transformer monitoring unit **100** showing connection details between cable assemblies and the connectors **118**, **120**, **124** of the transformer monitoring unit **100**. For example, a primary side current sensing cable assembly **402** may connect to the first connector **124**, a secondary side current sensing cable assembly **408** may connect to the second connector **118**, and a power cable assembly **414** may connect to the third connector **120**, when such third connector **120** is included. The primary current sensing cable assembly **402** may include a cable connector **404**, which mates with the first connector **124** of the transformer monitoring unit **100**, and a single conductor or multi-conductor cable **406** that connects to one more Rogowski coils or current transformers for sensing current at the primary side of the transformer. Similarly, the secondary current sensing cable assembly **408** may include a cable connector **410**, which mates with the second connector **118** of the transformer monitoring unit **100**, and a single conductor or multi-conductor cable **412** that is connected to one or more Rogowski coils or current transformers for sensing current at one or more secondary sides of the transformer.

When used, the power cable assembly **414** includes a cable connector **416**, which mates with the third connector **120** of the transformer monitoring unit **100**, and a single conductor or multi-conductor cable **418** for connecting to the secondary side of the transformer and providing power to the transformer monitoring unit **100**. The connectors **404**, **410**, **416** of the cable assemblies **402**, **408**, **414** may be Bayonet Neill-Concelman (BNC) type coaxial connectors that include grooves or slots, which engage respective lugs on complementary BNC connectors **118**, **120**, **124** of the transformer monitoring unit **100**. As illustrated in FIG. 4, the first connector **124** of the transformer monitoring unit **100** orients the primary current sensing cable assembly **402** such that the cable **406** of the assembly **402** extends in a direction generally opposite from that of the cable **412** of the secondary current sensing cable assembly **408** (and the power cable **418**) when the connector **410** of the secondary current sensing cable assembly **408** is connected to the second connector **118** of the transformer monitoring unit **100**. Such an arrangement helps prevent the primary and secondary current sensing cables **406**, **412** from crossing, overlapping, or otherwise being in close proximity to each other.

(20) FIG. 5 is a front perspective view of the exemplary transformer monitoring unit **100** as connected to various cable assemblies for use in monitoring an aerial pole-mounted transformer. In general, the transformer monitoring unit **100** can be used for monitoring aerial and pad mounted transformers. In FIG. 5, the transformer monitoring unit **100** is shown inverted relative to how it is shown in FIGS. 1-4. This orientation may be selected when the primary bushing(s) of the transformer is located at a higher position than the secondary bushing(s) of the transformer. In this embodiment, the primary current sensing cable assembly **402** includes or is connected to at least one Rogowski coil **508** (one shown) or current transformer that will be disposed around a bushing or other input on the primary side of the transformer. When positioned around a conductor, a Rogowski coil outputs a signal representative of an electric current passing through the conductor by electromagnetic coupling to the magnetic field produced by such current. Thus, Rogowski coil **508** is positioned around a conductor passing into the primary side of the transformer and generates output signals representative of the primary input current to the transformer (e.g., voltage signals that are proportional to such current over time). Similarly, the secondary current sensing cable assembly **408** includes or is connected to two Rogowski coils **502**, **506** in this embodiment, which are each positioned around a respective conductor on the secondary side of the transformer and output signals representative of the secondary output currents of the transformer. When used, power cable assembly **414** includes three terminals **510**, **512**, **514**, which are each connected to different voltage lines out of the secondary side of the transformer (e.g., hot, neutral, ground) to provide power to the transformer monitoring unit **100**. A pressure sensor cable **516** may be connected to the pressure sensor connector **306** on the back side **300** of the monitoring unit's housing **102**, and to a pressure sensor connected to a pressure sensor output of the transformer.

(21) FIG. 6 illustrates a use case of the exemplary transformer monitoring unit **100** as mounted to an aerial distribution transformer **602**. A pole **604** is typically used to suspend high voltage distribution lines above the ground, and the distribution transformer **602** is used to step down the voltage level from the primary distribution voltage level (e.g., 10 kV) to a service level (e.g., 110-480 V) that is provided to residences and businesses. The distribution lines are connected to the primary inputs **606**, **608** of the transformer **602** and stepped down by the transformer **602** to provide service level voltages at the secondary outputs **610**, **612** of the transformer **602**. The transformer monitoring unit **100** is substantially identical to that shown in FIG. 5. The primary side Rogowski coil **508** is shown positioned around one of the primary side conductors (e.g., conductor bushing) **608** sourcing current into the transformer **602**. On the secondary side, Rogowski coils **502**, **506** are positioned around secondary conductors (e.g., conductor bushings) **610**, **612**. Power cable conductors **510**, **512**, **514** may be connected to the secondary conductors **610**, **612**, **614**. Pressure sensor cable **516** may be connected to a pressure sensor attached to a pressure sensor output of the transformer **602**. The arrangement shown in FIG. 6 allows the transformer monitoring

unit **100** to sense and measure current at the primary and secondary sides of the transformer **602**, as well as to monitor the secondary output voltages of the transformer **602** and the pressure inside the oil tank of the transformer **602**. In a further exemplary embodiment, the transformer monitoring unit **100** may include communication circuitry to enable the transformer monitoring unit **100** to communicate monitored parameter data to a remote computing system. The transformer monitoring unit **100** is shown mounted on the outside wall of the transformer **602** and the configuration of the transformer monitoring unit **100** and its connectors **118**, **120**, **124**, with primary and secondary side connectors **118**, **124** located proximate opposite ends of the monitoring unit's housing **102** maintains physical separation between the cables **406**, **412** of the primary and secondary current sensing cable assemblies **402**, **408**.

(22) FIG. 7 is a front perspective view of the exemplary transformer monitoring unit **100** as connected to various cable assemblies for use in monitoring a pad mounted distribution transformer. As shown in the figure, the transformer monitoring unit **100** is oriented as shown in FIGS. 1-4, with the first connector **124** at the bottom and the second connector **118** at the top. As described above, the first and second connectors **118**, **124** may be multi-pin connectors allowing for different numbers of input signal lines. Accordingly, in the configuration shown in FIG. 7, the primary current sensing cable assembly **402** includes or is attached two Rogowski coils **702**, **704** or current transformers and the secondary current sensing cable assembly **408** also includes or is attached to two Rogowski coils **706**, **708** or current transformers. The power cable assembly **414** includes conductors **710**, **712**, **714** for connecting to the conductors of the secondary side of the transformer. A pressure sensor coupling **716** is configured to connect the pressure sensor cable **720** to a pressure sensor in the transformer. A pressure sensor cable **720** is further coupled to the pressure sensor connector **306** on the back side **300** of the transformer monitoring unit's housing **102**. An antenna coupling **718** is connected to the transformer monitoring unit **100** via an antenna cable **722**, which is connected to the antenna connector **304** on the back side **300** of the transformer monitoring unit's housing **102**.

(23) FIG. 8 illustrates a use case of the exemplary transformer monitoring unit **100** as mounted to a pad mounted distribution transformer **804**. The pad mounted transformer **804** may be mounted on a concrete pad or slab **802**. The transformer **804** illustrated in FIG. 8 is shown with its front or hatch door removed for clarity. The hatch door would be included to keep the conductors (bushings) protected from environmental conditions and isolated from contact by unauthorized personnel.

(24) In this embodiment, the transformer monitoring unit **100** may be secured to the wall separating the transformer tank from the transformer hatch. The transformer monitoring unit **100** is substantially identical to that shown in FIG. 7. The primary side conductors (bushings) **806**, **808** receive the distribution voltage, which is stepped down to service voltage levels at the secondary conductors (bushings) **812**, **814**, **818**. To sense and measure primary input current to one primary side conductor **806** and primary output from another primary side conductor **808** (e.g., to pass the primary distribution line to the primary input of another pad mounted distribution transformer), two Rogowski coils **702**, **704** or current transformers are coupled around the primary conductors **806**, **808** and connected to the transformer monitoring unit **100** by the primary current sensing cable assembly **402**. To sense and measure secondary output currents from secondary side conductors **812**, **818**, two Rogowski coils **706**, **708** or current transformers are positioned around the secondary conductors **818**, **812** and connected to the transformer monitoring unit **100** by the secondary current sensing cable assembly **406**. Power conductors **710**, **712**, **714** are coupled to the secondary conductors **818**, **812**, **814** and connected to the transformer monitoring unit **100** by the power cable assembly **414**. Pressure sensor **716** is connected to the pressure sensor connector **306** of the transformer monitoring unit **100** via a pressure sensor cable **720**, and an external antenna **718** is coupled to the antenna connector **304** of the transformer monitoring unit **100** via an antenna cable **722**.

(25) The arrangement shown in FIG. 8 allows the transformer monitoring unit **100** to sense and

measure current at the primary and secondary sides of the transformer **804**, as well as to monitor the secondary output voltages of the transformer **804** and the pressure inside the oil tank of the transformer **804**. In a further exemplary embodiment, the transformer monitoring unit **100** may include communication circuitry to enable the transformer monitoring unit **100** to communicate monitored parameter data to a remote computing system.

(26) As illustrated in FIGS. **7** and **8**, the first connector **124** of the transformer monitoring unit **100** orients the primary current sensing cable assembly **402** such that the cable **406** of the assembly **402** extends in a direction generally opposite from the direction of the cable **412** of the secondary current sensing cable assembly **408** when the connector **410** of the secondary current sensing cable assembly **408** is connected to the second connector **118** of the transformer monitoring unit **100**. Such an arrangement helps prevent the primary and secondary current sensing cables **406**, **412** from crossing, overlapping, or otherwise being in close proximity to each other, thereby mitigating errors in primary and/or secondary current values determined by the transformer monitoring unit **100**.

(27) The claims appended hereto are meant to cover all modifications and changes within the scope and spirit of the present disclosure.

Claims

1. A monitoring apparatus for a transformer, the monitoring apparatus comprising: a housing having a first end and a second end, wherein the second end is substantially opposite the first end; a first connector positioned proximate the first end of the housing, the first connector being configured for connection to a first cable assembly that outputs signals representative of at least one primary input current to the transformer; and a second connector positioned proximate the second end of the housing, the second connector being configured for connection to a second cable assembly that outputs signals representative of at least one secondary output current of the transformer; wherein the first connector and the second connector are oriented such that the first cable assembly extends away from the first end of housing in a first direction when the first cable assembly is connected to the first connector and the second cable assembly extends away from the second end of the housing in a second direction when the second cable assembly is connected to the second connector, and wherein the first direction and the second direction are generally opposite directions.
2. The monitoring apparatus of claim 1, wherein: the first connector is oriented at an angle of not more than forty-five degrees relative to a reference plane running a length of the housing; and the second connector is oriented at an angle of not less than one hundred thirty-five degrees relative to the reference plane.
3. The monitoring apparatus of claim 1, wherein: the second connector is oriented at an angle of not more than forty-five degrees relative to a reference plane running a length of the housing; and the first connector is oriented at an angle of not less than one hundred thirty-five degrees relative to the reference plane.
4. The monitoring apparatus of claim 1, wherein the first connector and the second connector are oriented so as to be at an angle in a range of about ninety degrees to about one hundred eighty degrees relative to each other.
5. The monitoring apparatus of claim 1, wherein each of the first connector and the second connector includes a respective connector component that extends outward from the housing.
6. The monitoring apparatus of claim 1, wherein the housing includes a base on a back side thereof, wherein the base defines a cavity, and wherein the monitoring apparatus further comprises: a pressure sensor connector positioned in the cavity; and an antenna connector positioned in the cavity.
7. The monitoring apparatus of claim 6, wherein a wall of the base defines at least two cutouts to

allow for passage of an antenna cable to the antenna connector and a pressure sensor cable to the pressure sensor connector.

8. The monitoring apparatus of claim 1, wherein the first cable assembly outputs signals representative of at least two primary input currents to the transformer.

9. The monitoring apparatus of claim 1, wherein the second cable assembly outputs signals representative of at least two secondary output currents of the transformer.

10. The monitoring apparatus of claim 1, further comprising: a third connector positioned proximate the second end of the housing, the third connector being configured for connection to a third cable assembly that outputs alternating current power for use by the monitoring apparatus.

11. The monitoring apparatus of claim 10, wherein the third connector is oriented substantially parallel to the second connector.

12. The monitoring apparatus of claim 1, wherein the first connector and the second connector are coaxial connectors.

13. The monitoring apparatus of claim 1, wherein the housing has a front side and a back side and wherein the first connector and the second connector are positioned on the front side of the housing.

14. The monitoring apparatus of claim 1, wherein the first end of the housing is a first lengthwise end of the housing and wherein the second end of the housing is a second lengthwise end of the housing.

15. A monitoring apparatus for a transformer, the monitoring apparatus comprising: a housing having a front side, a back side, a first lengthwise end, and a second lengthwise end, the second lengthwise end being substantially opposite the first lengthwise end; a first connector positioned on the front side of the housing proximate the first lengthwise end, the first connector being configured for connection to a first cable assembly that outputs signals representative of at least one primary input current to the transformer; and a second connector positioned on the front side of the housing proximate the second lengthwise end, the second connector being configured for connection to a second cable assembly that outputs signals representative of at least one secondary output current of the transformer; and a third connector positioned on the front side of the housing proximate the second lengthwise end, the third connector being configured for connection to a third cable assembly that outputs alternating current power for use by the monitoring apparatus; wherein the first connector and the second connector are oriented such that the first cable assembly extends away from the first end of housing in a first direction when the first cable assembly is connected to the first connector, the second cable assembly extends away from the second end of the housing in a second direction when the second cable assembly is connected to the second connector, and the first connector and the second connector are oriented so as to be at an angle in a range of about ninety degrees to about one hundred eighty degrees relative to each other.

16. The monitoring apparatus of claim 15, wherein each of the first connector and the second connector includes a respective connector component that extends outward from the front side of the housing.

17. The monitoring apparatus of claim 15, wherein the housing includes a base on the back side thereof, wherein the base defines a cavity, and wherein the monitoring apparatus further comprises: a pressure sensor connector positioned in the cavity; and an antenna connector positioned in the cavity.

18. The monitoring apparatus of claim 17, wherein a wall of the base defines at least two cutouts to allow for passage of an antenna cable to the antenna connector and a pressure sensor cable to the pressure sensor connector.

19. The monitoring apparatus of claim 15, wherein the first cable assembly outputs signals representative of at least two primary input currents to the transformer.

20. The monitoring apparatus of claim 15, wherein the second cable assembly outputs signals representative of at least two secondary output currents of the transformer.

